

IVAN MAKARENKO

TECHNICAL PRODUCT MANAGER · ROBOTICS ENGINEER · HARDWARE & MANUFACTURING

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Robotics engineer with technical product management experience in hardware development, supplier evaluation, product validation, and automated manufacturing. Direct experience working with customers and cross-functional teams across product, engineering, quality, sourcing, and manufacturing.

EXPERIENCE

SourceOne

Guangdong, China

Technical Product Management Intern

Jun - Aug 2026

- Represented SourceOne in evaluations of approximately 10 large automated factories across China, working with engineering, quality, and sourcing leaders to assess production capability, process controls, manufacturability, and supplier risk.
- Supported product validation and supplier reviews for Mainspring Energy, SKAN, Rough Country, and other customers. Worked directly with customer and factory engineers to review project issues, define next steps, test products, and examine quality documentation.
- Prepared supplier and product evaluation reports summarizing customer requirements, factory findings, quality and process risks, and follow-up actions for product, engineering, and sourcing teams.

Stantec

Chandler, AZ

Automation Engineering Intern

Jun - Aug 2024

- Identified errors in approximately 25% of legacy electrical schematics and updated Revit models across 12 mining infrastructure projects, improving drawing accuracy and helping prevent coordination issues during design reviews.
- Standardized Revit drawing templates for complex electrical systems and built an Excel project tracker that centralized documentation across electrical, mechanical, and civil teams, improving consistency and project visibility.

Addictive Desert Designs

Mesa, AZ

New Product Development Intern

Jun - Aug 2023

- Built a vehicle trim and fitment database covering 50+ truck models, helping shorten design cycles by approximately 30% and accelerate development of new off-road components.
- Developed off-road components from vehicle laser scanning and SolidWorks CAD through robotic assembly and testing, then partnered with OTC engineers to commission two robotic welding systems for expanded production capacity.

Freelance Consulting Practice

Remote

Founder & IT Systems Consultant

Jun 2019 - Present

- Built a referral-based IT consulting practice serving 20+ clients and generating \$20K-\$70K in annual revenue, with no client churn over five years while studying full-time.
- Led client discovery, requirements definition, prioritization, delivery, and iteration for custom software, websites, and analytics tools across construction, legal services, and retail.

SELECTED PROJECTS

FoldRide Pro

New York, NY

Inventor & Product Lead

2025 - Present

- Designed a foldable electric skateboard with a structural spine lock and a truck system that can be removed without tools. Translated user needs into product requirements, system architecture, engineering drawings, and factory-ready STEP files.
- Filed a U.S. provisional patent application covering the folding architecture and truck release system. Worked with factory engineers to improve manufacturability, safety systems, the dual motor drive, and backpack integration toward a retail target below \$999.

Robotic Driving System for NASA Psyche Mission

Mesa, AZ

ASU Senior Capstone

Jan - Dec 2024

- Led mechanical and electrical subsystem design for an asteroid rover, placing approximately 70% of its mass in the wheels and developing independent suspension that enabled 360-degree turns without reversing in near-zero gravity and metallic dust conditions.
- Worked with NASA mission mentors to replace hydraulic actuation with a spring system compatible with vacuum conditions, then delivered a prototype that met weight, size, speed, and budget requirements.

EDUCATION

Columbia University

New York, NY

M.S. in Technology Management, School of Professional Studies

Expected Dec 2026

Focus: Product Management, AI Management, and Digital Transformation

Arizona State University

Mesa, AZ

B.S. in Engineering, Concentration in Robotics, Ira A. Fulton Schools of Engineering

Dec 2024

Focus: Electrical Engineering and Mechatronics

SKILLS

Product: Product strategy · Roadmapping · Requirements definition · Customer discovery · Product validation · Supplier evaluation

Engineering: DFM · Robotic automation · Embedded systems · PCB design · Product testing · Quality analysis

Tools: Python · C/C++ · SQL · MATLAB · SolidWorks · Revit · AutoCAD · Cadence · Git · Excel

Languages: English (Fluent) · Russian (Fluent) · Ukrainian (Fluent)